



Course Prepared by
IITian & AI Architect

Duration: 8 Months
6 Months Learning + 2 Months Internship

AI ENGINEERING

Learn AI. Build Intelligent Applications. Engineer Real-World AI Systems.

SKILL IT Education presents a structured, practical career pathway in AI Engineering — taking learners step-by-step from Python foundations and backend development to Machine Learning, Generative AI, LLM application development, Agentic AI, and production-grade AI systems. Every concept is reinforced through hands-on projects, real-world tools and industry-aligned engineering practices.

Artificial Intelligence has become the core driver of modern software, automation and business decision-making. Organizations across every industry now need engineers who can design, build, deploy and maintain AI-powered systems — making AI Engineering one of the fastest-growing and most valuable career paths of this decade.

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Program Overview

Build Your AI Engineering Career

This program takes you step-by-step from strong foundations in Python to building intelligent applications with Machine Learning, Generative AI, LLMs, Agentic AI — and finally deploying production-ready AI systems for real business use cases.

Why This Matters

AI now drives decision-making, automation and product innovation across every industry. Engineers who can design, build and deploy AI systems — not just use AI tools — are in the highest demand across the global technology job market.

*From Learning to Building. From Projects to Production.
From Skills to Career.*

1

Python & Technical Foundations

2

Python Backend Development with AI

3

Machine Learning Engineering

4

Generative AI & LLM Application Engineering

5

Agentic AI Engineering

6

MLOps, LLMOps & AI Platform Engineering

7

AI Solutions Engineering

Module 1 — Python & Technical Foundations

WHAT YOU WILL LEARN

- Python Programming
- Functions & Object-Oriented Programming
- Error Handling
- Virtual Environments
- Git & GitHub Version Control
- Linux / Command Line Interface
- Working with JSON
- HTTP & REST APIs
- Debugging Techniques
- Basic Testing
- Technical Documentation

TOOLS & TECHNOLOGIES

Python

VS Code

Git

GitHub

Linux

Postman

REST APIs

JSON



Learning Outcome: Students build a strong software foundation for AI development by writing Python applications, working with APIs, managing code with Git/GitHub, debugging applications and applying professional development practices.

04 Module 2 — Python Backend Development with AI

WHAT YOU WILL LEARN

- Advanced Python & Modular Application Architecture
- FastAPI Development & RESTful API Design
- PostgreSQL & MongoDB Database Fundamentals
- Redis & Background Job Queues
- Authentication, Authorization & JWT / OAuth
- API Security & Role-Based Access Control
- Testing & Test-Driven Development
- Docker Containerization
- CI/CD with GitHub Actions
- Cloud Deployment
- Capstone: Production-Style Backend Service

TOOLS & TECHNOLOGIES

Python

FastAPI

PostgreSQL

MongoDB

Redis

Docker

GitHub Actions

Pytest

AWS

Azure



Learning Outcome: Students build robust, secure and scalable backend systems with Python and FastAPI — working with relational and NoSQL databases, implementing authentication and API security, and deploying production-ready services with Docker, CI/CD and the cloud. This module lays the essential backend foundation for every AI engineering role ahead.

WHAT YOU WILL LEARN

- Machine Learning Fundamentals
- Supervised Learning
- Unsupervised Learning
- Feature Engineering
- Model Training & Evaluation
- ML Pipelines
- Deep Learning Fundamentals
- Model Serving & Deployment
- Model Monitoring & Retraining

TOOLS & TECHNOLOGIES

Python

NumPy

Pandas

Scikit-learn

PyTorch

Jupyter

FastAPI

MLflow

Docker

ML ENGINEERING PIPELINE

Data

Features

Model

Eval

API

Docker

Monitor



Learning Outcome: Students learn to develop, evaluate, package, serve and monitor Machine Learning models, and to build practical, production-ready ML services end to end.

Module 4 — Generative AI & LLM Application Engineering

WHAT YOU WILL LEARN

- Generative AI Fundamentals
- Large Language Models
- LLM APIs & Prompt Engineering
- Structured Outputs & Function Calling
- Embeddings & Vector Databases
- Document Ingestion & Chunking
- Retrieval-Augmented Generation (RAG)
- Reranking & RAG Evaluation
- PostgreSQL & pgvector
- MCP Fundamentals

TOOLS & TECHNOLOGIES

LLM APIs

Hugging Face

LangChain

PostgreSQL

pgvector

Vector DBs

FastAPI

Docker

MCP

RAG APPLICATION FLOW

Docs

Chunk

Embed

Vector DB

Retrieve

LLM

Answer



Learning Outcome: Students learn to build LLM-powered applications, integrate AI models with external data and APIs, develop RAG systems, implement vector search and ship practical Generative AI applications.

Module 5 — Agentic AI Engineering

WHAT YOU WILL LEARN

- AI Agent Fundamentals & Architecture
- Tool Calling & Function Calling
- State & Memory Management
- Routing & Agent Workflows
- Model Context Protocol (MCP)
- Multi-Agent Patterns
- Human-in-the-Loop Design
- Retries, Fallbacks & Guardrails
- Agent Evaluation & Observability
- Prompt Injection Defence

TOOLS & TECHNOLOGIES

Python

LangGraph

MCP

LLM APIs

FastAPI

PostgreSQL

Vector DBs

Docker

AGENT ARCHITECTURE

User

Agent

Reasoning

MCP/APIs

Action

Evaluate



Learning Outcome: Students design, build and evaluate AI agents that call tools, maintain state, execute multi-step workflows, and operate safely and reliably in production.

Module 6 — MLOps, LLMOps & AI Platform Engineering

WHAT YOU WILL LEARN

- Experiment Tracking & Model Registry
- Model & Prompt Versioning
- AI & RAG Evaluation
- Regression Testing
- CI/CD for AI Systems
- Tracing & Observability
- Cloud IAM & Secrets Management
- Scaling & Rollbacks
- Cost & Latency Management
- AI Security & Reliability

TOOLS & TECHNOLOGIES

MLflow

Git

GitHub Actions

Docker

Cloud Platform

PostgreSQL / pgvector

CI/CD

Monitoring Tools

AI PLATFORM LIFECYCLE

Develop

Test

Evaluate

Deploy

Monitor

Improve



Learning Outcome: Students learn to operationalize AI systems end-to-end — tracking experiments, versioning models and prompts, automating testing and deployment, and monitoring production AI for cost, latency, security and reliability.

Module 7 — AI Solutions Engineering

WHAT YOU WILL LEARN

- Client Discovery & Requirement Gathering
- Workflow Mapping
- AI Solution Architecture
- Data & System Integration
- API Integration
- Security & PII Handling
- Solution Design & Prototyping
- UAT & Deployment
- Cost Estimation & Technical Documentation
- Architecture Diagrams & Client Demonstrations

TOOLS & TECHNOLOGIES

Python

FastAPI

REST APIs

PostgreSQL

LLM APIs

RAG

MCP

Docker

Cloud Platform

Git/GitHub

SOLUTION DEVELOPMENT JOURNEY

Problem

Reqs

Arch.

Prototype

Integration

UAT

Deploy

Demo



Learning Outcome: Students learn to gather requirements, design AI solution architectures, integrate systems and APIs, and prototype and deploy client-ready AI solutions with proper documentation and demonstration.

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AI Engineering Technology Stack

The complete toolset used across every module of the program

Programming

Python

Development

Git

GitHub

Linux

VS Code

Backend & APIs

FastAPI

REST APIs

PostgreSQL

MongoDB

Data & Retrieval

pgvector

Vector Databases

Redis

Machine Learning

Scikit-learn

PyTorch

Generative AI

LLM APIs

Hugging Face

LangChain

RAG

Agent Engineering

LangGraph

MCP

Tool Calling

Tracing

DevOps & Testing

Docker

GitHub Actions

Pytest

Cloud Platforms

AWS

Azure

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Project-Based Learning

Build What You Learn — every module is reinforced with hands-on work

01

Python AI Application

Build Python applications using APIs, data processing pipelines and automation scripts.

02

Backend API Service

Build and deploy a secure, production-style REST API with FastAPI, PostgreSQL/MongoDB, authentication and Docker.

03

Machine Learning Service

Train, evaluate and deploy Machine Learning models as production-ready services.

04

Generative AI / RAG Application

Build LLM-powered apps with RAG, vector search and document understanding.

05

Agentic AI Application

Build intelligent agents that use tools, make decisions and execute multi-step tasks.

FINAL CAPSTONE

End-to-End AI Solution

Design and build a complete AI solution from requirements to deployment and live demonstration.

Learning Outcomes

What You Will Be Able to Build by the End of the Program

- ✓ Develop AI applications using Python
- ✓ Build and deploy secure backend APIs & services
- ✓ Build and evaluate Machine Learning models
- ✓ Develop deep learning solutions
- ✓ Build LLM-powered applications
- ✓ Develop Retrieval-Augmented Generation (RAG) systems
- ✓ Work with embeddings and vector databases
- ✓ Build and orchestrate AI agents
- ✓ Integrate external tools and APIs
- ✓ Work with LangGraph and MCP
- ✓ Deploy AI applications to production
- ✓ Evaluate and monitor AI systems
- ✓ Apply AI security and reliability practices
- ✓ Design end-to-end AI solution architectures
- ✓ Build, present and demonstrate real AI projects

Backend & Software Engineering

- Backend Developer
- API Developer
- Python Developer
- Software Engineer

AI & Machine Learning

- AI Engineer
- Machine Learning Engineer
- MLOps Engineer
- Machine Learning Developer

Generative AI

- Generative AI Engineer
- GenAI Application Engineer
- LLM Application Developer
- AI Application Developer

Agentic AI

- Agentic AI Engineer
- AI Agent Developer
- AI Automation Engineer
- AI Workflow Developer

AI Production & Platform

- MLOps Engineer
- LLMOps Engineer
- AI Platform Engineer
- AI Infrastructure Engineer

AI Solutions

- AI Solutions Engineer
- AI Integration Engineer
- AI Implementation Engineer
- AI Consultant

Technical Assessment & Industry Certifications

Measure your skills. Improve your performance.

Weekly Assessments

Regular technical assessments to measure understanding of recently completed topics.

Module-End Assessments

Comprehensive technical evaluation after completion of each major module.

Practical / Coding Assessments

Students demonstrate their ability to implement concepts through coding and real-world tasks.

Project Assessments

Evaluate project architecture, implementation, testing, documentation and presentation.

CERTIFICATION PATHWAYS ALIGNED WITH COURSE CONTENT

Microsoft Azure AI Engineer

AWS ML Engineer / Cloud Practitioner

Google Cloud Professional ML Engineer

NVIDIA Generative AI & LLMs

Databricks ML / GenAI Engineer

Oracle Cloud Infrastructure AI

IBM AI / Generative AI Engineering

TensorFlow & Hugging Face Credentials

Certification Notice

The curriculum is aligned to help students prepare for these external certifications. Registration, fees and eligibility for any external exam are managed directly by the respective certifying body and may change over time. SKILL IT Education awards its own course-completion certificate upon successful completion of the program.

Internship, Industry Exposure & Placement Support

Take your skills into practice — from learning to career

INTERNSHIP & INDUSTRY EXPOSURE

After successful course completion and applicable project evaluations, eligible students can progress toward internship and practical industry exposure opportunities, subject to availability and selection criteria.

AI Application Dev.	Machine Learning
Generative AI / LLM	RAG
Agentic AI	AI Automation
API Integration	AI Deployment

Internship Journey



PLACEMENT SUPPORT

- Resume preparation guidance
- LinkedIn & GitHub portfolio guidance
- Project presentation preparation
- Technical interview preparation
- Mock interviews & coding practice
- HR interview & communication prep
- Job search & interview opportunity sharing
- Interview feedback & improvement guidance

Career Readiness Journey





WHY CHOOSE SKILL IT EDUCATION

DON'T JUST LEARN AI. LEARN TO ENGINEER IT.

Learn. Build. Deploy. Evaluate. Demonstrate.

SKILL IT Education builds AI engineers, not just AI users. Through an industry-driven curriculum, project-based learning, real-world tools and hands-on capstone work, every student graduates with a portfolio of production-style AI systems, verified skills and the confidence to walk into technical interviews and deliver from day one. This is where practical learning meets genuine career growth — in Machine Learning, Generative AI, Agentic AI and beyond.

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Let's build the future with AI.

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